

Matthew Kim

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EDUCATION

Bachelor of Applied Science (BASc), Mechatronics Engineering (Candidate)

University of Waterloo

Expected June 2030

Waterloo, ON

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks & Libraries: React, Next.js, Hono, Tailwind CSS, Flask, FastAPI, REST APIs, NetworkX

Tools: Git, GitHub, Docker, Vite, PostgreSQL, PostGIS, Redis, Supabase, OpenAI API, WCAG/ARIA, CAN/LIN

EXPERIENCE

Software Developer

Jan 2026 – Present

University of Waterloo Libraries

Waterloo, ON

- Rebuild a legacy Drupal 8 Assignment Planner into a standalone web application using TypeScript, HTML, CSS, and Hono, supporting ~3,000 daily active users (GA4).
- Re-architect CMS-coupled logic into an independent application, eliminating upgrade-related breakage and improving long-term maintainability for a university-wide student resource.
- Apply WCAG/ARIA accessibility standards and use Docker + Git/GitHub to support consistent environments and collaborative development workflows.

Firmware Developer

Sep 2025 – Dec 2025

Formula Electric – University of Waterloo

Waterloo, ON

- Developed embedded C/C++ firmware for ECUs managing vehicle communication via CAN/LIN networks and multiple sensor interfaces.
- Implemented HIL testing, signal simulation, and fault diagnostics, reducing per-board validation time by 35%.
- Collaborated with electrical and controls teams to debug hardware and improve real-time signal timing.

Engineering Research Assistant

Feb 2024 – Jun 2024

FAST Research Group – Western University

London, ON

- Designed 15+ CAD components in Fusion 360 and produced 10+ FDM prints using Cura, supporting prototyping for an open-source multi-material 3D food printer.
- Performed mechanical testing, calibration, and iterative refinements, increasing print consistency by 25%.
- Conducted experimental data collection and analysis, evaluating crop and photovoltaic performance in varying conditions.

PROJECTS

WATWay

Oct 2025 – Present

TypeScript, Python, Next.js, FastAPI, PostgreSQL, PostGIS, Redis, Docker, OpenAI API

- Develop a full-stack campus navigation platform using Next.js and FastAPI, delivering indoor and accessibility-aware routes unavailable in Google/Apple Maps.
- Implement backend routing logic (A*) supporting fastest, accessible, and weather-aware paths using PostgreSQL/PostGIS, NetworkX, and Redis caching.
- Integrate calendar data and AI-assisted queries via OpenAI APIs to generate personalized route recommendations.

Pantry Pal

Nov 2025

TypeScript, React, Vite, Tailwind CSS, Supabase, Edamam API

- Fide x Lovable Hackathon Winner – built an AI-powered kitchen assistant that analyzes pantry photos, identifies ingredients, and recommends recipes via the Edamam API.
- Developed a TypeScript-based recipe matching system in React that ranks meals based on ingredient coverage.
- Built a drag-and-drop meal planner and shopping list using React DnD with Supabase-backed persistence to streamline weekly planning and grocery management for users.

QueueMe

May 2025

TypeScript, Python, React, Flask, REST APIs, Supabase

- Built a full-stack queue-management web app using React, Flask, REST APIs, and Supabase, enabling real-time line tracking for 100+ simulated users.
- Implemented backend endpoints to automate join/leave operations, improving responsiveness and user experience.
- Integrated Supabase for persistent storage and live status updates, maintaining stable behaviour under concurrent usage.